Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 10

Roll No : 496

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01) WAP to read and display the contents of a text file. (also try to open the file in some other directory)

- in the form of a string

- line by line

- in the form of a list

In [2]:
```
fp = open("same_directory.txt" , "r")
print(fp.read())
fp.seek(0)
print(fp.readline())
fp.seek(0)
print(fp.readlines()) #returns list of lines with \n at last
fp.close()

fp = open("../diff_directory.txt" , "r")
print(fp.read())
fp.seek(0)
print(fp.readline())
fp.seek(0)
print(fp.readlines()) #returns list of lines with \n at last
fp.close()
```

```
File in same directory
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```

02) WAP to create file named "new.txt" only if it doesn't exist.

In [2]:
```
fp = open("new2.txt", "x")
fp.close()
```

03) WAP to read first 5 lines from the text file.

In [7]:
```
fp = open("same_directory.txt" , "r")
for _ in range(5):
    print(fp.readline() ,end="")
fp.close()
```

```
File in same directory
Python language is easy to Learn.
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```

04) WAP to find the longest word(s) in a file.

In [12]:
```
fp=open('new2.txt','r')
text=fp.read()
fp.close()
words=text.split()
max_length=0
for i in words:
    if len(i)>max_length:
        max_length=len(i)
for i in words:
    if len(i)==max_length:
        print(i)
```

```
particularly
```

05) WAP to count the no. of lines, words and characters in a given text file.

In [37]:
```
fp = open("new2.txt" , "r")

lines = fp.readlines()
fp.close()
print("Total lines :" , len(lines))

word_count = 0
char_count = 0
for i in lines:
    words = i.split()
    word_count  += len(words)
    char_count += len(i)

print("Total words :",word_count)
print("Total characters :",char_count)
```

```
Total lines : 10
Total words : 166
Total characters : 1090
```

06) WAP to copy the content of a file to the another file.

In [26]:
```
fp = open("new2.txt" , "r")
fp1 = open("same_directory.txt" , "w")
fp1.write(fp.read())
fp1.close()
fp.close()
```

07) WAP to find the size of the text file.

In [41]:
```
fp = open("new2.txt" , "r")
fp.seek(0,2)
print(fp.tell())
fp.close()
```

```
1090
```

08) WAP to create an UDF named frequency to count occurances of the specific word in a given text file.

In [44]:
```python
def frequency(search_word):
    fp = open('new2.txt', "r")
    text = fp.read()
    fp.close()

    words = text.split()
    count = 0

    for word in words:
        if word == search_word:
            count += 1

    return count


word = input("Enter the word to search: ")

occurrences = frequency(word)
print("Occurrences of", word, ":", occurrences)
```

```
Occurrences of Python : 8
```

09) WAP to get the score of five subjects from the user, store them in a file. Fetch those marks and find the highest score.

In [46]:
```python
fp = open("marks.txt", "w")

for i in range(5):
    mark = int(input("Enter marks of subject " + str(i+1) + ": "))
    fp.write(str(mark) + "\n")

fp.close()

fp = open("marks.txt", "r")
```

```
marks = []
for line in fp:
    marks.append(int(line.strip()))

fp.close()

highest = marks[0]
for m in marks:
    if m > highest:
        highest = m

print("Highest score:", highest)
```

```
Highest score: 98
```

10) WAP to write first 100 prime numbers to a file named primenumbers.txt

(Note: each number should be in new line)

In [47]:
```
fp = open("primenumbers.txt", "w")

count = 0
num = 2

while count < 100:
    is_prime = True

    for i in range(2, int(num//2) + 1):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        fp.write(str(num) + "\n")
        count += 1

    num += 1

fp.close()
```

11) WAP to merge two files and write it in a new file.

In [48]:
```
fp1 = open("new2.txt", "r")
fp2 = open("same_directory.txt", "r")
fp3 = open("merged.txt", "w")

data1 = fp1.read()
data2 = fp2.read()

fp3.write(data1)
fp3.write(data2)

fp1.close()
fp2.close()
fp3.close()
```

12) WAP to replace word1 by word2 of a text file. Write the updated data to new file.

In [3]:
```
fp = open("new2.txt", "r")

data = fp.read()

word1 = input("Enter word to replace: ")
word2 = input("Enter new word: ")

new_data = data.replace(word1, word2)

fp.close()

# write updated content to new file
fp2 = open("output12.txt", "w")
fp2.write(new_data)
fp2.close()

print("Word replaced successfully and saved in output.txt")
```

```
Word replaced successfully and saved in output.txt
```

13) Demonstrate tell() and seek() for all the cases(seek from beginning-end-current position) taking a suitable example of your choice.

In [9]:
```
f = open("new2.txt", "rb")

# initial position
print("Initial position:", f.tell())

# read first 5 bytes
data = f.read(5)
print("Read first 5 bytes:", data)

# position after reading
print("Position after reading:", f.tell())

# move pointer to beginning
f.seek(0, 0)
print("Position after seek(0,0):", f.tell())
```

```python
# read again
print("Read again:", f.read(5))

# move 2 bytes forward from current position
f.seek(2, 1)
print("Position after seek(2,1):", f.tell())

# move 5 bytes before end
f.seek(-5, 2)
print("Position near end:", f.tell())

print("Last bytes:", f.read())

f.close()
```

```
Initial position: 0
Read first 5 bytes: b'Pytho'
Position after reading: 5
Position after seek(0,0): 0
Read again: b'Pytho'
Position after seek(2,1): 7
Position near end: 1085
Last bytes: b'ates.'
```

In [ ]: